What is Fine-Tuning?

100

→ Start with small Context length
≃ 2K or 4K,

CL: 8K, 12K / Domain-specific data

Pre-training → mid-training → SFT → Instruct tuning → RLHF

Constitutional AI

tokens >> 1T

≃ 1B

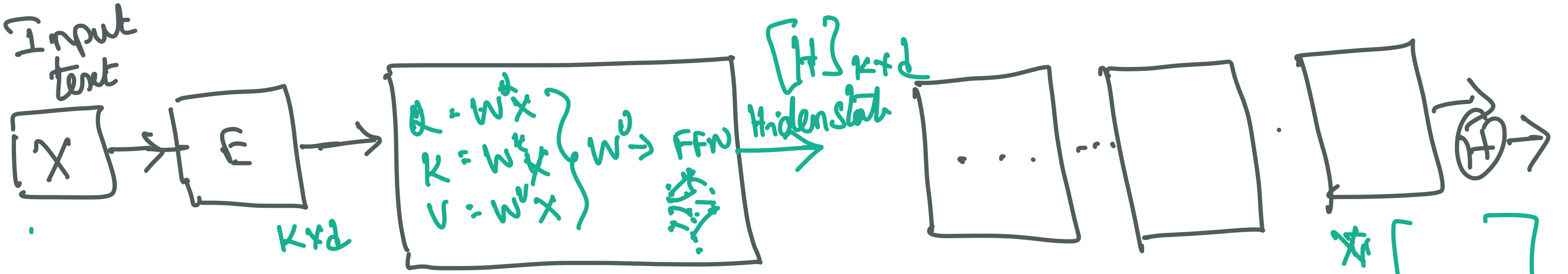
Checkpoint 1 CP2 CP3 CP4 CP5 CP6 CP7 CP8 CPn . . . .

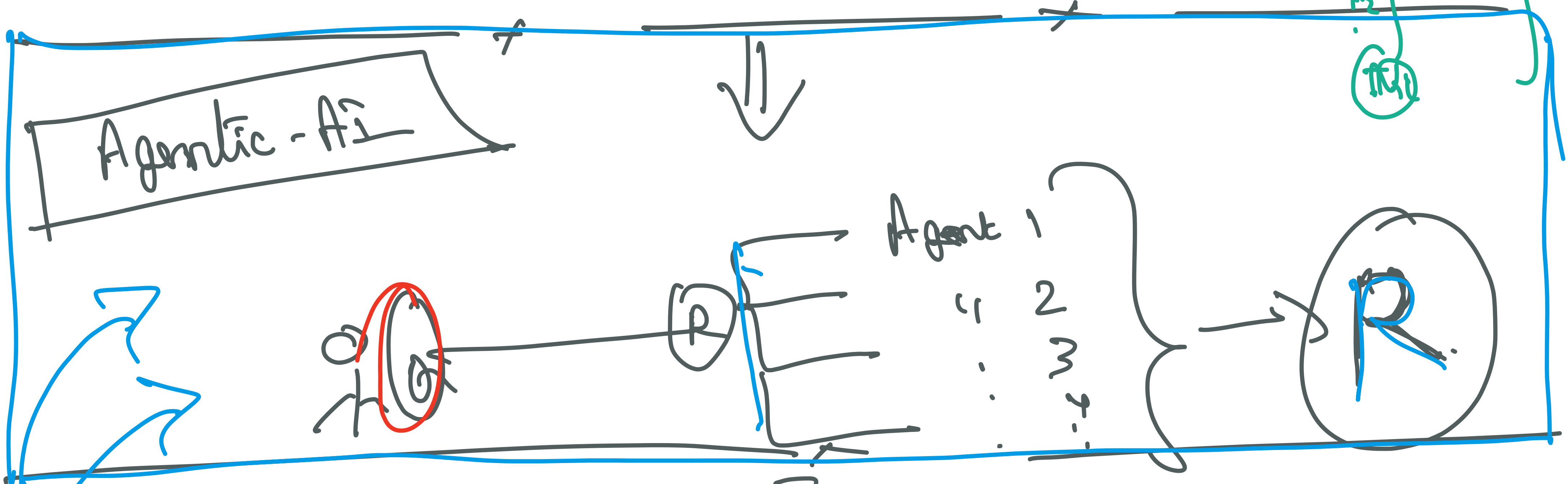
Continued training that permanently encodes new behaviour into model weights

Base model → LLM

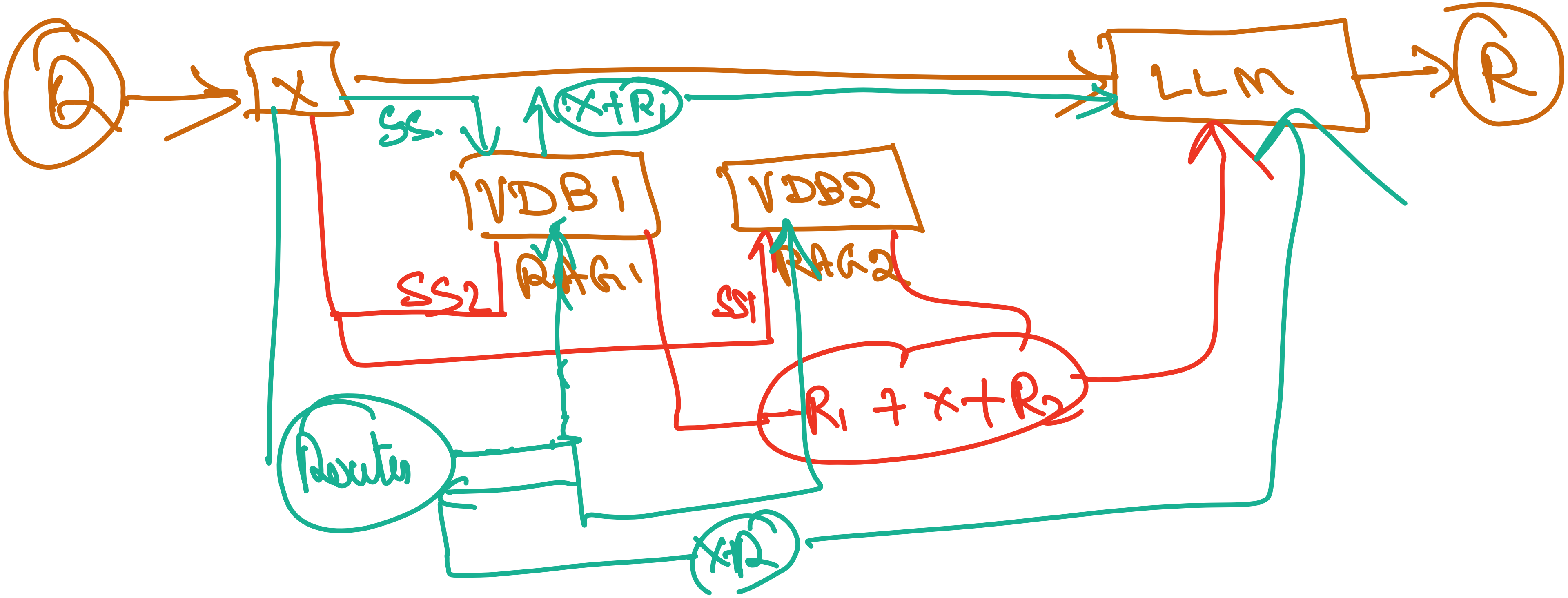
8 min	B2 – Understand	L2 – Intermediate	AI-FT-IN-TH-000001
Duration	Bloom's	Level	ID

I like Sum





1. Prompting [System]
2. RAG ← [addition knowledge to the model]



Q
X
SS
X+R1
LLM
R
VDB1
VDB2
SS2
RAG1
SS1
RAG2
R1 + X + R2
Router
X+R2

Fine-Tuning: The One-Sentence Definition

Definition

Fine-tuning is the continued training of a pre-trained foundation model on a curated, task-specific dataset, resulting in updated model weights that permanently encode the new behaviour — producing a $\boxed{\text{distinct model checkpoint.}}$

US law ← C_{100} → Indian law FT → C_{101}

- Unlike prompting: no weight changes occur — the model is unchanged after a prompt
- Unlike RAG: no external retrieval store is needed — knowledge is baked into weights
- The result is a new artefact: a fine-tuned checkpoint different from the base model
- Fine-tuning is persistent — the adapted behaviour survives across all future calls
- Real examples: GPT-4o (instruction-tuned), Med-PaLM (domain-specialist), CodeLlama (code)

The 3-Strategy Adaptation Landscape

Strategy 1
Prompting

Add examples or instructions to the context window. No code changes, no cost.

Best for: General tasks, one-off queries, exploration

→ Insufficient? Add knowledge →

Strategy 2
RAG

Retrieve relevant documents and inject into the prompt. Knowledge stays fresh.

Best for: Factual Q&A, up-to-date knowledge, large corpora

from vector DB

relative

→ Need behaviour change? →

Strategy 3
Fine-Tuning

Train on task-specific data to update weights. Persistent, efficient at inference.

Best for: Specific behaviour, tone, domain expertise

Prompting vs RAG vs Fine-Tuning — Side by Side

Dimension	Prompting	RAG	Fine-Tuning
Weight change?	None	None	Yes — permanent
Setup cost	None	Index + retriever	Data + training
Inference overhead	Token cost only	Retrieval latency	None (merged weights)
Knowledge update	Instant	Add to index	Re-train required
Behaviour shaping	Limited	Limited	Strong
Example	Chain-of-thought	Enterprise Q&A	CodeLlama, Med-PaLM

Anatomy of a Fine-Tuned Checkpoint

1

Base Model

Pre-trained on trillions of tokens. General language capability.

2

Fine-Tuning Data

Task-specific instruction–response pairs. Hundreds to millions of examples.

3

Training Run

Gradient updates computed on task loss. Weights adjusted.

4

Fine-Tuned Checkpoint

New .bin / .safetensors file. Distinct artefact from base model.

5

Deploy

Replace base model at inference. No extra prompt overhead.

Real-World Fine-Tuned Model Taxonomy

Category	What is tuned	Example Models	Use Case
Instruction-Tuned	Follow instructions precisely	GPT-4o, Claude 3, Llama 3 Instruct	General assistants
Domain-Specialist	Deep domain knowledge	Med-PaLM 2, CodeLlama, LegalBERT	Medical, code, legal
Style-Adapted	Tone, voice, format	Brand voice bots, formal/casual tone	Customer service
Safety-Aligned	Refusal, harmlessness (RLHF)	GPT-4, Claude, Gemini	Consumer products
Task-Specialist	Single narrow task	T5-summarisation, Whisper	Summarisation, ASR

When Fine-Tuning Is the Right Choice — 5 Signals

✅ Behaviour must be consistent

You need the model to always respond in a specific format, tone, or style — not just sometimes.

✅ Large labelled dataset available

You have hundreds+ of (instruction, output) pairs specific to your task.

✅ Prompt engineering is insufficient

Few-shot examples in the context don't achieve the quality you need.

✅ Latency / cost matters at scale

A fine-tuned smaller model can match a large prompted model at lower per-call cost.

⚠️ RAG is insufficient

The task requires the model to reason in a domain-specific way, not just retrieve facts.

⚠️ Start with prompting first

Always try prompting → RAG → fine-tune. Fine-tuning has upfront cost.

KEY REFERENCES

- Wei, J. et al. (2021). Finetuned Language Models Are Zero-Shot Learners. ICLR 2022.
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- Jurafsky & Martin (2025). Speech and Language Processing, Ch. 10. web.stanford.edu/~jurafsky/slp3

▶ **Next: Concept 2: Full-Parameter Fine-Tuning (AI-FT-IN-TH-000002)**